Brutal Practice Paper B29 - Acids/Bases & Salts | Algebraic Expressions | Wastewater Story | Integers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. Blue litmus turns red in liquid X, while liquid Y turns red litmus blue. To neutralise 1 cup of X you need 4 cups of Y. How many cups of Y neutralise 7 cups of X, and what colour does turmeric paper show in the neutral mixture? [Acids, Bases & Salts]

- **(A)** 28 cups; turmeric turns red
- **(B)** 4 cups; turmeric stays yellow
- **(C)** 28 cups; turmeric stays yellow
- **(D)** 32 cups; turmeric stays yellow
- **(E)** 28 cups; turmeric turns blue
- **(F)** 11 cups; turmeric stays yellow
- **(G)** 7 cups; turmeric turns red

2. If a = 3 and b = -2, find the value of 2a^2 - 3ab + b^2. [Algebraic Expressions]

- **(A)** -40
- **(B)** 58
- **(C)** 40
- **(D)** 36
- **(E)** 0
- **(F)** 22

3. A treatment plant receives 12,000 litres of sewage. Bar screens and grit removal take out solids that are 1/8 of the volume. How many litres move on to the sedimentation tank? [Wastewater Story]

- **(A)** 9,600 litres
- **(B)** 10,000 litres
- **(C)** 11,500 litres
- **(D)** 1,500 litres
- **(E)** 8,000 litres
- **(F)** 13,500 litres
- **(G)** 10,500 litres

4. A submarine is at -180 m. It rises 25 m each minute for 4 minutes, then dives 60 m. What is its final depth? [Integers]

- **(A)** -80 m
- **(B)** -100 m
- **(C)** -40 m
- **(D)** -220 m
- **(E)** -120 m
- **(F)** -140 m
- **(G)** -145 m

5. One antacid tablet neutralises 5 mL of stomach acid and costs Rs 4. A patient secretes 40 mL of excess acid. What is the least cost to neutralise all of it, and the antacid works because it is a what? [Acids, Bases & Salts]

- **(A)** Rs 8; a base
- **(B)** Rs 160; a base
- **(C)** Rs 32; a salt
- **(D)** Rs 32; an acid
- **(E)** Rs 40; a base
- **(F)** Rs 32; a base

6. Subtract 3x^2 - 5x + 2 from 7x^2 - 2x - 4, then find the value of the result when x = 2. [Algebraic Expressions]

- **(A)** 4
- **(B)** 10
- **(C)** 16
- **(D)** 22
- **(E)** -16
- **(F)** 18
- **(G)** 28

7. Put these treatment steps in correct early-to-late order: aeration, bar screens, sedimentation, grit removal. Where do floating scum and settling sludge first separate? [Wastewater Story]

- **(A)** aeration, sedimentation, grit removal, bar screens; at sedimentation
- **(B)** bar screens, grit removal, aeration, sedimentation; at aeration
- **(C)** bar screens, grit removal, sedimentation, aeration; at sedimentation
- **(D)** grit removal, bar screens, aeration, sedimentation; at aeration
- **(E)** sedimentation, bar screens, grit removal, aeration; at sedimentation
- **(F)** bar screens, grit removal, sedimentation, aeration; at aeration
- **(G)** bar screens, sedimentation, grit removal, aeration; at sedimentation

8. Evaluate (-8) + (-12) / 4 x 3 - (-5) using the correct order of operations. [Integers]

- **(A)** -30
- **(B)** -22
- **(C)** 6
- **(D)** -2
- **(E)** -18
- **(F)** -12
- **(G)** 2

9. A student lists six kitchen liquids: lemon juice, soap solution, vinegar, lime water, table-salt solution, and curd. How many are acidic, and what fraction of the six is acidic? [Acids, Bases & Salts]

- **(A)** 4 acidic; two-thirds
- **(B)** 2 acidic; one-half
- **(C)** 3 acidic; one-half
- **(D)** 3 acidic; one-third
- **(E)** 4 acidic; one-half
- **(F)** 3 acidic; one-fourth
- **(G)** 2 acidic; one-third

10. A rectangle has length (2x + 3) cm and width (x - 1) cm. Write its perimeter in simplest form and find it when x = 5. [Algebraic Expressions]

- **(A)** 6x + 2; 32 cm
- **(B)** 2x + 2; 12 cm
- **(C)** 6x + 4; 26 cm
- **(D)** 6x + 4; 30 cm
- **(E)** 6x + 8; 38 cm
- **(F)** 6x + 4; 34 cm
- **(G)** 3x + 2; 17 cm

11. In an aeration tank, helpful bacteria use oxygen to eat wastes. Air is pumped for 6 hours per batch, and the plant runs 4 batches a day. How many hours of aeration daily, and these oxygen-using bacteria are called? [Wastewater Story]

- **(A)** 24 hours; aerobic bacteria
- **(B)** 2 hours; aerobic bacteria
- **(C)** 10 hours; aerobic bacteria
- **(D)** 48 hours; aerobic bacteria
- **(E)** 24 hours; anaerobic bacteria
- **(F)** 24 hours; harmful bacteria
- **(G)** 12 hours; aerobic bacteria

12. In a quiz, +4 is given for a correct answer, -2 for a wrong one, and 0 for a skip. Ravi attempts 25 questions: 15 correct, 7 wrong, 3 skipped. What is his score? [Integers]

- **(A)** 74
- **(B)** 52
- **(C)** 48
- **(D)** 40
- **(E)** 46
- **(F)** 32
- **(G)** 60

13. Equal-strength acid and base neutralise each other millilitre for millilitre. A student adds 30 mL of acid to 50 mL of this base. What is left unreacted, and how does the final mixture affect red litmus? [Acids, Bases & Salts]

- **(A)** 20 mL of base; red litmus turns red
- **(B)** 20 mL of base; red litmus stays red
- **(C)** 20 mL of acid; red litmus turns blue
- **(D)** 30 mL of base; red litmus turns blue
- **(E)** 20 mL of base; red litmus turns blue
- **(F)** 80 mL of salt; no change

14. Add the three expressions 3a - 2b + 5, -a + 4b - 7, and 2a - b + 3. Then find the value when a = 1 and b = 2. [Algebraic Expressions]

- **(A)** 4a + b + 1; value 5
- **(B)** 4a + 5b + 1; value 15
- **(C)** 4a + b + 1; value 7
- **(D)** 4a + b + 1; value 11
- **(E)** 4a - b + 1; value 3
- **(F)** 6a + b + 1; value 9
- **(G)** 4a + b - 1; value 5

15. Sludge is sent to a digester where anaerobic bacteria produce biogas. If 500 kg of sludge gives 30 cubic metres of biogas, how much biogas comes from 1.5 tonnes of sludge, and the gas is used as? [Wastewater Story]

- **(A)** 90 cubic metres; fuel
- **(B)** 60 cubic metres; fuel
- **(C)** 90 cubic metres; fertiliser only
- **(D)** 900 cubic metres; fuel
- **(E)** 30 cubic metres; fuel
- **(F)** 45 cubic metres; fuel
- **(G)** 90 cubic metres; drinking gas

16. A bank account starts at -350 rupees (overdrawn). Four deposits of 200 each are made, then 600 is withdrawn. What is the final balance? [Integers]

- **(A)** -150 rupees
- **(B)** -1150 rupees
- **(C)** 450 rupees
- **(D)** 150 rupees
- **(E)** -450 rupees
- **(F)** 50 rupees
- **(G)** -250 rupees

17. A bee sting is acidic but a wasp sting is basic. A child suffers 4 bee stings and 3 wasp stings. Bee stings are rubbed with baking soda; wasp stings with vinegar. How many vinegar rubs are needed, and vinegar is a mild what? [Acids, Bases & Salts]

- **(A)** 3 vinegar rubs; a salt
- **(B)** 3 vinegar rubs; a mild acid
- **(C)** 4 vinegar rubs; a mild acid
- **(D)** 7 vinegar rubs; a mild acid
- **(E)** 3 vinegar rubs; a strong acid
- **(F)** 3 vinegar rubs; a mild base
- **(G)** 4 vinegar rubs; a mild base

18. In the term $-7x^2y^3$, what is the numerical coefficient, and what is the sum of the powers of the variables (the degree)? [Algebraic Expressions]

- **(A)** 7; degree 6
- **(B)** -7; degree 6
- **(C)** 7; degree 5
- **(D)** -7; degree 5
- **(E)** $-7x^2$; degree 5
- **(F)** -7; degree 2

19. A city of 50,000 people produces 120 litres of sewage per person per day. How many litres of sewage per day, and untreated, this water mainly threatens health by spreading? [Wastewater Story]

- **(A)** 600,000 litres; waterborne diseases
- **(B)** 60,000 litres; waterborne diseases
- **(C)** 6,000,000 litres; air pollution
- **(D)** 6,000,000 litres; waterborne diseases
- **(E)** 5,000,000 litres; waterborne diseases
- **(F)** 60,000,000 litres; waterborne diseases
- **(G)** 6,000,000 litres; noise pollution

20. Find the value of (-5) x (-3) x (-2) x 4. [Integers]

- **(A)** 60
- **(B)** -40
- **(C)** -120
- **(D)** -100
- **(E)** -30
- **(F)** -60
- **(G)** 120

21. Acidic soil is treated with lime (a base) spread at 2 kg for every 5 square metres. A farmer must treat 220 square metres. How much lime is needed, and lime is added because the soil is too what? [Acids, Bases & Salts]

- **(A)** 88 kg; too salty
- **(B)** 80 kg; too acidic
- **(C)** 44 kg; too acidic
- **(D)** 88 kg; too acidic
- **(E)** 440 kg; too acidic
- **(F)** 88 kg; too basic
- **(G)** 110 kg; too acidic

22. Ravi's age is 3 years more than twice Meera's age. Meera is x years old. Write Ravi's age 5 years ago as an expression, and find it when Meera is 10. [Algebraic Expressions]

- **(A)** 2x + 8; 28 years
- **(B)** 2x - 2; 18 years
- **(C)** 2x + 2; 22 years
- **(D)** 2x - 2; 23 years
- **(E)** 2x - 2; 20 years
- **(F)** 2x - 8; 12 years
- **(G)** x - 2; 8 years

23. Treated water leaving a plant is 95% clean of contaminants. If the raw sewage carried 800 mg of contaminants per litre, how much remains per litre after treatment? [Wastewater Story]

- **(A)** 95 mg per litre
- **(B)** 760 mg per litre
- **(C)** 400 mg per litre
- **(D)** 8 mg per litre
- **(E)** 80 mg per litre
- **(F)** 20 mg per litre
- **(G)** 40 mg per litre

24. The temperature at 6 am was -4 degC and rose 3 degC each hour until noon. What was the temperature at noon? [Integers]

- **(A)** 2 degC
- **(B)** 14 degC
- **(C)** 18 degC
- **(D)** 20 degC
- **(E)** 12 degC
- **(F)** -14 degC
- **(G)** 8 degC

25. China rose indicator turns dark pink in acids and green in bases. A student tests five samples and records: pink, green, pink, pink, green. How many samples are acidic, and what percent of the five is acidic? [Acids, Bases & Salts]

- **(A)** 3 acidic; 65%
- **(B)** 2 acidic; 60%
- **(C)** 2 acidic; 40%
- **(D)** 3 acidic; 30%
- **(E)** 3 acidic; 50%
- **(F)** 4 acidic; 80%
- **(G)** 3 acidic; 60%

26. A pattern uses 2n + 1 matchsticks to make n triangles in a row. How many matchsticks are needed for 15 triangles, and which term first uses more than 40 matchsticks? [Algebraic Expressions]

- **(A)** 31 sticks; the 40th
- **(B)** 30 sticks; the 20th
- **(C)** 16 sticks; the 20th
- **(D)** 31 sticks; the 20th
- **(E)** 31 sticks; the 21st
- **(F)** 45 sticks; the 20th
- **(G)** 31 sticks; the 19th

27. From this list - cooking oil, used motor oil, paints, fruit peels meant for compost, and old medicines - how many should be kept OUT of household drains, and why mainly? [Wastewater Story]

- **(A)** 2 items; they kill microbes
- **(B)** 4 items; they make water taste bad
- **(C)** 4 items; they add oxygen
- **(D)** 4 items; they clog pipes or kill helpful microbes
- **(E)** 1 item; it clogs pipes
- **(F)** 5 items; they clog pipes
- **(G)** 3 items; they clog pipes or kill microbes

28. Find the sum of all integers from -6 to 9, both included. [Integers]

- **(A)** 3
- **(B)** 30
- **(C)** -24
- **(D)** 45
- **(E)** 21
- **(F)** 15
- **(G)** 24

29. When hydrochloric acid completely reacts with sodium hydroxide, it forms a salt and water and gives out heat. If 3 units of this acid need 3 units of the base, what is needed for 9 units of acid, and is heat taken in or given out? [Acids, Bases & Salts]

- **(A)** 3 units of base; heat given out
- **(B)** 9 units of base; no heat change
- **(C)** 9 units of base; heat taken in
- **(D)** 6 units of base; heat given out
- **(E)** 9 units of base; heat given out
- **(F)** 9 units of salt; heat taken in
- **(G)** 27 units of base; heat given out

30. Simplify 5(2x - 3) - 2(3x - 4) and then find its value at x = -1. [Algebraic Expressions]

- **(A)** 10x - 7; value -17
- **(B)** 4x - 7; value -3
- **(C)** 4x - 7; value 3
- **(D)** 4x + 1; value -3
- **(E)** 4x - 7; value -11
- **(F)** 4x - 23; value -27

31. A septic tank for a remote house fills with sludge and needs emptying every 3 years. Over 18 years, how many times is it emptied, and a septic tank is an example of what kind of system? [Wastewater Story]

- **(A)** 6 times; a river-cleaning system
- **(B)** 6 times; on-site sanitation
- **(C)** 3 times; on-site sanitation
- **(D)** 5 times; on-site sanitation
- **(E)** 6 times; a large sewage plant
- **(F)** 9 times; on-site sanitation
- **(G)** 6 times; open drainage

32. A diver descends 12 m every minute from the surface (0 m). What is her depth after 7 minutes, and then she ascends 30 m - what is the new depth? [Integers]

- **(A)** -84 m then 54 m
- **(B)** -84 m then -54 m
- **(C)** 84 m then 54 m
- **(D)** -54 m then -84 m
- **(E)** -72 m then -42 m
- **(F)** -19 m then 11 m
- **(G)** -84 m then -114 m

33. Milk of magnesia (magnesium hydroxide) is taken for acidity. One dose neutralises 6 mL of excess stomach acid. A patient has 30 mL of excess acid. How many doses clear it, and milk of magnesia is a what? [Acids, Bases & Salts]

- **(A)** 4 doses; a base
- **(B)** 5 doses; a neutral substance
- **(C)** 5 doses; a salt
- **(D)** 6 doses; a base
- **(E)** 5 doses; a base
- **(F)** 5 doses; an acid
- **(G)** 10 doses; a base

34. The cost of p pens at Rs 8 each and q notebooks at Rs 15 each is written as an expression. Find the total cost when p = 6 and q = 4, after a flat Rs 10 discount. [Algebraic Expressions]

- **(A)** 8p + 15q - 10; Rs 108
- **(B)** 8p + 15q - 10; Rs 98
- **(C)** 8p + 15q - 10; Rs 90
- **(D)** 23pq - 10; Rs 542
- **(E)** 8p + 15q - 10; Rs 88
- **(F)** 8p + 15q + 10; Rs 118
- **(G)** 15p + 8q - 10; Rs 112

35. Chlorine is added to clear water to kill remaining germs. A plant uses 4 g of chlorine per 1000 litres. For 25,000 litres, how much chlorine is needed, and chlorination is the last main step to make water? [Wastewater Story]

- **(A)** 25 g; disinfected
- **(B)** 100 g; aerated
- **(C)** 100 g; disinfected
- **(D)** 100 g; muddier
- **(E)** 1000 g; disinfected
- **(F)** 100 g; salty
- **(G)** 40 g; disinfected

36. Evaluate (-48) / (-6) / (-2). [Integers]

- **(A)** -4
- **(B)** 4
- **(C)** -16
- **(D)** 16
- **(E)** 1
- **(F)** -1
- **(G)** -8

37. Phenolphthalein is colourless in acid and pink in base. A flask of base is pink. Acid is added drop by drop until the pink just vanishes, using 12 drops. To reach the same end-point the next day needed 3 fewer drops. How many drops the second day, and at the end-point the solution is? [Acids, Bases & Salts]

- **(A)** 15 drops; neutral
- **(B)** 9 drops; basic
- **(C)** 12 drops; neutral
- **(D)** 9 drops; neutral
- **(E)** 3 drops; neutral
- **(F)** 9 drops; still pink

38. From the sum of 4x^2 + 3x - 7 and -2x^2 + 5, subtract x^2 - 2x + 1. Give the simplified expression. [Algebraic Expressions]

- **(A)** x^2 + 3x - 3
- **(B)** x^2 + 5x - 5
- **(C)** x^2 + 5x - 1
- **(D)** x^2 + x - 3
- **(E)** 3x^2 + 5x - 3
- **(F)** x^2 - 5x - 3
- **(G)** x^2 + 5x - 3

39. A treatment plant cleans 8,00,000 litres a day in summer and 1/4 more in the monsoon. How many litres a day does it clean in the monsoon? [Wastewater Story]

- **(A)** 10,00,000 litres
- **(B)** 8,00,000 litres
- **(C)** 6,00,000 litres
- **(D)** 2,00,000 litres
- **(E)** 11,00,000 litres
- **(F)** 12,00,000 litres

40. A lift starts at floor +8. It goes down 11 floors, up 4 floors, then down 6 floors. Which floor does it reach (negative means basement)? [Integers]

- **(A)** 5 (5th floor)
- **(B)** -3
- **(C)** 7
- **(D)** -5 (5th basement)
- **(E)** -9
- **(F)** 3
- **(G)** -1

41. A gardener has soil that turns red litmus blue. To fix it she should add an acidic material. From sawdust-compost (acidic), wood ash (basic), and quicklime (basic), how many suitable materials are there, and the soil was originally what? [Acids, Bases & Salts]

- **(A)** 0 suitable; basic soil
- **(B)** 1 suitable; basic soil
- **(C)** 1 suitable; acidic soil
- **(D)** 1 suitable; neutral soil
- **(E)** 2 suitable; acidic soil
- **(F)** 3 suitable; basic soil
- **(G)** 2 suitable; basic soil

42. A taxi charges Rs 50 plus Rs 12 per km. Write the fare for d km, and find how many km give a fare of Rs 230. [Algebraic Expressions]

- **(A)** 62d; 4 km
- **(B)** 50 + 12d; 15 km
- **(C)** 50 + 12d; 18 km
- **(D)** 50 + 12d; 19 km
- **(E)** 12 + 50d; 4 km
- **(F)** 50 + 12d; 16 km

43. In sedimentation, solids settle as sludge while lighter scum floats. If 10,000 L of water yields 250 L of sludge and 50 L of scum, what fraction of the water became sludge? [Wastewater Story]

- **(A)** 1/4
- **(B)** 1/100
- **(C)** 1/20
- **(D)** 1/200
- **(E)** 1/50
- **(F)** 1/40
- **(G)** 3/100

44. The product of two integers is -36. If one of them is -4, what is the other, and is the pair's sum positive or negative? [Integers]

- **(A)** -9; sum is positive
- **(B)** 9; sum is positive
- **(C)** 9; sum is zero
- **(D)** -9; sum is negative
- **(E)** 9; sum is negative
- **(F)** 12; sum is positive
- **(G)** -12; sum is negative

45. Vinegar is 5% acetic acid by volume. A bottle holds 500 mL of vinegar. How many millilitres of pure acetic acid does it contain, and acetic acid turns blue litmus to what? [Acids, Bases & Salts]

- **(A)** 25 mL; red
- **(B)** 2.5 mL; red
- **(C)** 50 mL; red
- **(D)** 100 mL; red
- **(E)** 250 mL; red
- **(F)** 25 mL; blue
- **(G)** 25 mL; stays blue

46. A number is x. Three more than four times the number, decreased by the number itself, is taken. Write the simplified expression and find its value when x = 9. [Algebraic Expressions]

- **(A)** 3x + 3; value 33
- **(B)** 3x + 3; value 30
- **(C)** 5x + 3; value 48
- **(D)** 3x + 6; value 33
- **(E)** 3x - 3; value 24
- **(F)** 4x + 3; value 39

47. A village builds toilets so cases of a waterborne disease fall from 80 a year to 16 a year. By what percent did the cases fall? [Wastewater Story]

- **(A)** 64%
- **(B)** 20%
- **(C)** 84%
- **(D)** 80%
- **(E)** 60%
- **(F)** 75%
- **(G)** 16%

48. A shopkeeper records daily profit or loss for a week (in rupees): +120, -50, +80, -90, +40, -30, +60. What is the net result for the week? [Integers]

- **(A)** +260
- **(B)** +130 (profit)
- **(C)** -70
- **(D)** -130
- **(E)** +70
- **(F)** +470
- **(G)** +150

49. Three test tubes hold an acid, a base, and a neutral solution. Red litmus is dipped in all three; then blue litmus is dipped in all three. In total, how many of these six dips show a colour change? [Acids, Bases & Salts]

- **(A)** 6 dips
- **(B)** 0 dips
- **(C)** 2 dips
- **(D)** 5 dips
- **(E)** 3 dips
- **(F)** 4 dips
- **(G)** 1 dip

50. How many terms are in 5x^2 - 3xy + 2y^2 - 7, and how many of them are like the term -4xy? [Algebraic Expressions]

- **(A)** 5 terms; 1 like term
- **(B)** 4 terms; 2 like terms
- **(C)** 7 terms; 1 like term
- **(D)** 4 terms; 1 like term
- **(E)** 3 terms; 1 like term
- **(F)** 4 terms; 3 like terms
- **(G)** 4 terms; 0 like terms

51. A sewer pipe slopes downward by 1 metre for every 400 metres of length. For a 2 km sewer, what is the total drop, and the wastewater flows mainly because of? [Wastewater Story]

- **(A)** 5 metres; gravity
- **(B)** 2 metres; gravity
- **(C)** 5 metres; wind
- **(D)** 5 metres; pumps only
- **(E)** 0.5 metres; gravity
- **(F)** 8 metres; gravity
- **(G)** 800 metres; gravity

52. Fill the blank: (-7) x ___ = 84. What is the missing integer, and is it positive or negative? [Integers]

- **(A)** -11 (negative)
- **(B)** 12 (negative)
- **(C)** -91
- **(D)** -12 (negative)
- **(E)** -77
- **(F)** 12 (positive)
- **(G)** 11 (negative)

53. Sodium chloride (common salt) is made from an acid and a base. Which pair is correct, and is the salt acidic, basic, or neutral? [Acids, Bases & Salts]

- **(A)** hydrochloric acid + sodium hydroxide; neutral
- **(B)** sulphuric acid + sodium hydroxide; neutral
- **(C)** hydrochloric acid + sodium hydroxide; basic
- **(D)** nitric acid + potassium hydroxide; neutral
- **(E)** acetic acid + sodium hydroxide; neutral
- **(F)** hydrochloric acid + sodium hydroxide; acidic
- **(G)** hydrochloric acid + calcium hydroxide; neutral

54. A square has side (3x - 2) units. Write its perimeter, and find x if the perimeter is 64 units. [Algebraic Expressions]

- **(A)** 12x - 8; x = 4
- **(B)** 12x - 8; x = 5
- **(C)** 12x + 8; x = 4.67
- **(D)** 12x - 2; x = 5.5
- **(E)** 3x - 2; x = 22
- **(F)** 4x - 8; x = 18
- **(G)** 12x - 8; x = 6

55. Dried sludge becomes nutrient-rich manure. A digester yields 80 kg of manure from each tonne of wet sludge. From 3.5 tonnes of wet sludge, how much manure, and this reuse is an example of? [Wastewater Story]

- **(A)** 28 kg; recycling waste
- **(B)** 350 kg; recycling waste
- **(C)** 280 kg; recycling waste into a resource
- **(D)** 280 kg; making it toxic
- **(E)** 2800 kg; recycling waste
- **(F)** 280 kg; wasting a resource
- **(G)** 240 kg; recycling waste

56. On a number line, a frog starts at 0, jumps +5, then -8, then +3, then -6. Where does it land, and how far is that from the start? [Integers]

- **(A)** -6; 0 units
- **(B)** -22; 22 units
- **(C)** -12; 12 units
- **(D)** 6; 6 units from start
- **(E)** -6; -6 units from start
- **(F)** 0; 0 units
- **(G)** -6; 6 units from start

57. Litmus is a dye taken from lichens. In a kit there are 60 litmus strips, and 2/5 are blue, the rest red. A test uses 8 blue and 6 red strips. How many blue strips remain? [Acids, Bases & Salts]

- **(A)** 20
- **(B)** 8
- **(C)** 12
- **(D)** 18
- **(E)** 16
- **(F)** 10

58. Evaluate 3p - 2q + pq when p = -2 and q = -3. [Algebraic Expressions]

- **(A)** -12
- **(B)** -18
- **(C)** -6
- **(D)** 0
- **(E)** 12
- **(F)** 6
- **(G)** 18

59. A bar screen catches large floating objects. In one hour it removes 18 plastic items and 12 cloth items from the sewage. What fraction of the removed items are plastic, and bar screens are which stage of treatment? [Wastewater Story]

- **(A)** 3/5; the first stage
- **(B)** 3/5; the aeration stage
- **(C)** 18/12; the first stage
- **(D)** 2/5; the first stage
- **(E)** 2/3; the first stage
- **(F)** 3/5; the last stage
- **(G)** 1/2; the first stage

60. Compute -15 - (-23) + (-9) - 11. [Integers]

- **(A)** 12
- **(B)** -12
- **(C)** -20
- **(D)** 6
- **(E)** -34
- **(F)** -2

61. Tartaric acid (in tamarind) and baking soda (a base) fizz to make a salt and a gas. A recipe mixes them in a 3:2 mass ratio. For 18 g of tartaric acid, how much baking soda is needed, and the fizzing shows what kind of change? [Acids, Bases & Salts]

- **(A)** 12 g; physical change
- **(B)** 27 g; chemical change
- **(C)** 9 g; chemical change
- **(D)** 12 g; chemical change
- **(E)** 12 g; no change
- **(F)** 6 g; chemical change

62. The sum of three consecutive integers is written using n for the smallest. Write the simplified sum, and find the integers if the sum is 72. [Algebraic Expressions]

- **(A)** 3n; 24, 25, 26
- **(B)** n + 3; 69, 70, 71
- **(C)** 3n + 6; 22, 23, 24
- **(D)** 3n + 3; 24, 25, 26
- **(E)** 3n + 2; 23, 24, 25
- **(F)** 3n + 3; 23, 24, 25
- **(G)** 3n + 3; 22, 23, 24

63. Wastewater carries nitrogen and phosphorus nutrients. A river sample has 6 mg/L of nitrogen against a safe limit of 4 mg/L. By how much does it exceed the limit, and excess nutrients cause what? [Wastewater Story]

- **(A)** 2 mg/L over; nothing happens
- **(B)** 2 mg/L over; algae that later use up the water's oxygen
- **(C)** 2 mg/L over; algae that add oxygen forever
- **(D)** 4 mg/L over; algae that use up oxygen
- **(E)** 1.5 mg/L over; algae that use up oxygen
- **(F)** 10 mg/L over; algae that use up oxygen
- **(G)** 2 mg/L over; the water becomes acidic

64. The highest point of a region is +1,340 m and the lowest is -260 m. What is the difference in height between them? [Integers]

- **(A)** 1,600 m
- **(B)** 1,340 m
- **(C)** 1,060 m
- **(D)** -1,600 m
- **(E)** 260 m
- **(F)** 1,080 m

65. An acid solution needs 4 drops of base to neutralise it. A second acid, twice as strong in the same volume, needs how many drops? Then both neutral mixtures are combined - the final mix is? [Acids, Bases & Salts]

- **(A)** 8 drops; acidic
- **(B)** 2 drops; neutral
- **(C)** 16 drops; neutral
- **(D)** 8 drops; cannot tell
- **(E)** 4 drops; neutral
- **(F)** 8 drops; basic
- **(G)** 8 drops; neutral

66. A shop sells tea at Rs x per cup. On Monday it sells 40 cups and on Tuesday 55 cups. Write the two-day takings as an expression and find it when x = 12, after paying Rs 80 for milk. [Algebraic Expressions]

- **(A)** 95x - 80; Rs 980
- **(B)** 95x + 80; Rs 1220
- **(C)** 95x - 80; Rs 1140
- **(D)** 95x - 80; Rs 1000
- **(E)** 40x + 55 - 80; Rs 455
- **(F)** 95x - 80; Rs 1060
- **(G)** 85x - 80; Rs 940

67. A composting (vermi) toilet uses worms to treat waste with no water. A school replaces 5 flush toilets that each use 8 litres per flush, flushed 60 times a day each. How many litres of water are saved per day? [Wastewater Story]

- **(A)** 1,200 litres
- **(B)** 300 litres
- **(C)** 2,400 litres
- **(D)** 480 litres
- **(E)** 240 litres
- **(F)** 24,000 litres
- **(G)** 4,800 litres

68. A team plays 20 games scoring +6 per win, -4 per loss, and 0 per draw. It draws 4 games and ends with a net score of +6. How many games did it win? [Integers]

- **(A)** 11
- **(B)** 9
- **(C)** 8
- **(D)** 10
- **(E)** 6
- **(F)** 5
- **(G)** 7

69. Carbon dioxide dissolves in water to form carbonic acid, a weak acid, so fizzy drinks turn blue litmus slightly red. If 9 of 12 tested fizzy drinks do this, what fraction fail the acid test? [Acids, Bases & Salts]

- **(A)** 1/2
- **(B)** 3/4
- **(C)** 1/4
- **(D)** 1/3
- **(E)** 1/12
- **(F)** 9/12

70. An expression added to 4x^2 - 2x + 3 gives 9x^2 + x - 2. Find that expression. [Algebraic Expressions]

- **(A)** 5x^2 + 3x - 1
- **(B)** 5x^2 + 3x + 5
- **(C)** 5x^2 - 3x - 5
- **(D)** 5x^2 - x - 1
- **(E)** 5x^2 + 3x - 5
- **(F)** 13x^2 + 3x - 5

71. A plant releases 70% of its treated water to a river and uses the rest for gardening. If it treats 9,00,000 L, how much is used for gardening? [Wastewater Story]

- **(A)** 2,10,000 litres
- **(B)** 1,80,000 litres
- **(C)** 3,00,000 litres
- **(D)** 2,70,000 litres
- **(E)** 6,30,000 litres
- **(F)** 7,00,000 litres
- **(G)** 90,000 litres

72. Evaluate (-2) x [5 + (-9)] - (-3) x 4. [Integers]

- **(A)** 4
- **(B)** -12
- **(C)** 32
- **(D)** -20
- **(E)** 8
- **(F)** -4
- **(G)** 20

73. Ant bites inject methanoic (formic) acid. A soothing cream containing a base treats 3 bites per gram. A picnic group has 27 ant bites. How many grams of cream are needed, and the cream relieves pain by doing what? [Acids, Bases & Salts]

- **(A)** 12 g; neutralising the acid
- **(B)** 81 g; neutralising the acid
- **(C)** 3 g; neutralising the acid
- **(D)** 9 g; neutralising the acid
- **(E)** 9 g; adding more acid
- **(F)** 6 g; neutralising the acid
- **(G)** 9 g; cooling only

74. A garden is a rectangle of length 3x metres and width 2x metres. A square flower-bed of side x metres sits inside it. Write the leftover garden area, and find it when x = 4. [Algebraic Expressions]

- **(A)** x^2; 16 square metres
- **(B)** 5x; 20 square metres
- **(C)** 5x^2; 20 square metres
- **(D)** 7x^2; 112 square metres
- **(E)** 6x^2; 96 square metres
- **(F)** 5x^2; 80 square metres
- **(G)** 5x^2; 40 square metres

75. Cholera, typhoid and hepatitis spread through dirty water. In a survey of 200 sick people, 1/5 had cholera, 35% had typhoid, and the rest had other illnesses. How many had other illnesses? [Wastewater Story]

- **(A)** 110
- **(B)** 40
- **(C)** 130
- **(D)** 90
- **(E)** 70
- **(F)** 60
- **(G)** 100

76. The sum of two integers is -3 and their product is -28. What are the two integers? [Integers]

- **(A)** 14 and -2
- **(B)** -4 and 7
- **(C)** 4 and 7
- **(D)** -28 and 1
- **(E)** 4 and -7
- **(F)** -4 and -7

77. A neutral solution is made by mixing equal-strength acid and base. A student plans to mix 24 mL acid with 24 mL base, but spills 1/3 of the acid before mixing. To stay neutral, how much base should now be used? [Acids, Bases & Salts]

- **(A)** 12 mL
- **(B)** 20 mL
- **(C)** 16 mL
- **(D)** 18 mL
- **(E)** 8 mL
- **(F)** 32 mL
- **(G)** 24 mL

78. If 3x - 7 = 2x + 5, find x, then evaluate x^2 - 4x. [Algebraic Expressions]

- **(A)** x = 12; value 108
- **(B)** x = 4; value 0
- **(C)** x = 2; value -4
- **(D)** x = 12; value 144
- **(E)** x = 12; value 96
- **(F)** x = -2; value 12
- **(G)** x = 12; value 60

79. At a six-stage plant, water spends these minutes at the stages: 10, 20, 90, 360, 40, 20. What is the total time the water spends, in hours? [Wastewater Story]

- **(A)** 9 hours 30 minutes
- **(B)** 8 hours
- **(C)** 540 hours
- **(D)** 6 hours
- **(E)** 7 hours 30 minutes
- **(F)** 10 hours
- **(G)** 9 hours

80. A thermometer reads -2 degC. It falls 3 times by 4 degC each, then rises once by 5 degC. What is the final reading? [Integers]

- **(A)** -9 degC
- **(B)** 9 degC
- **(C)** -19 degC
- **(D)** -7 degC
- **(E)** -5 degC
- **(F)** -14 degC

81. Some salts are acidic, some basic, some neutral. In a set of 20 salts, 1/4 are acidic and 2/5 are basic; the rest are neutral. How many neutral salts are there? [Acids, Bases & Salts]

- **(A)** 7
- **(B)** 13
- **(C)** 3
- **(D)** 8
- **(E)** 9
- **(F)** 6
- **(G)** 5

82. The number of tiles bordering a square pool with n tiles per side (corners shared) is 4n - 4. How many tiles for n = 9, and how many tiles is the '-4' removing? [Algebraic Expressions]

- **(A)** 32 tiles; the 4 sides
- **(B)** 32 tiles; 2 corners
- **(C)** 40 tiles; the 4 corners
- **(D)** 32 tiles; the 4 shared corners
- **(E)** 36 tiles; the 4 corners
- **(F)** 28 tiles; the 4 corners
- **(G)** 32 tiles; nothing

83. Greywater (from sinks and baths) is less dirty than blackwater (from toilets). A home produces 240 L of greywater and 60 L of blackwater daily. What is the ratio of greywater to total wastewater in simplest form? [Wastewater Story]

- **(A)** 1 : 5
- **(B)** 1 : 4
- **(C)** 4 : 5
- **(D)** 4 : 1
- **(E)** 3 : 5
- **(F)** 5 : 4
- **(G)** 2 : 3

84. Divide -560 by 8, then multiply the result by -3. [Integers]

- **(A)** 210
- **(B)** 630
- **(C)** -630
- **(D)** 240
- **(E)** -70
- **(F)** -210
- **(G)** 70

85. Turmeric paper turns red in a base. A drop of soap solution reddens it; a drop of lemon juice then turns the red patch back to yellow. A third drop of lemon on the yellow patch does what, and lemon is what? [Acids, Bases & Salts]

- **(A)** turns blue; lemon is an acid
- **(B)** stays yellow; lemon is an acid
- **(C)** turns red; lemon is an acid
- **(D)** stays yellow; lemon is a base
- **(E)** stays yellow; lemon is neutral
- **(F)** turns green; lemon is an acid

86. Evaluate 2x^3 - x^2 + 4 when x = -2. [Algebraic Expressions]

- **(A)** 0
- **(B)** -24
- **(C)** 8
- **(D)** 16
- **(E)** -12
- **(F)** -16

87. A pump lifts sewage from a sump. During a power cut, the pump stops for 15 minutes while sewage keeps flowing in at 600 litres per minute. How many litres back up in the sump during the cut? [Wastewater Story]

- **(A)** 6,000 litres
- **(B)** 36,000 litres
- **(C)** 9,000 litres
- **(D)** 900 litres
- **(E)** 90,000 litres
- **(F)** 1,500 litres
- **(G)** 45 litres

88. A cave floor is at -45 m. A rope ladder has rungs every 3 m starting from ground level (0 m). How many rungs are there from 0 m down to the floor, not counting the top at 0 m? [Integers]

- **(A)** 14
- **(B)** 16
- **(C)** 30
- **(D)** 45
- **(E)** 13
- **(F)** 48
- **(G)** 15

89. A factory neutralises acidic waste using lime. Each litre of waste needs 30 g of lime. The factory releases 2,500 litres a day. How many kilograms of lime are needed per day, and the treated water then moves closer to being? [Acids, Bases & Salts]

- **(A)** 750 kg; neutral
- **(B)** 75 kg; acidic
- **(C)** 37.5 kg; neutral
- **(D)** 75 kg; neutral
- **(E)** 75 kg; basic
- **(F)** 2,500 kg; neutral

90. Two expressions are 6a + 4 and 4a + 10. For what value of a are they equal, and what is that common value? [Algebraic Expressions]

- **(A)** a = 7; value 46
- **(B)** a = 3; value 13
- **(C)** a = 3; value 22
- **(D)** a = -3; value -14
- **(E)** a = 2; value 16
- **(F)** a = 3; value 26
- **(G)** a = 3; value 18

91. Manholes are placed along a sewer every 50 metres for cleaning access. A straight sewer is 1.2 km long. How many manholes are needed if there is one at each end too? [Wastewater Story]

- **(A)** 24
- **(B)** 50
- **(C)** 23
- **(D)** 26
- **(E)** 25
- **(F)** 12

92. If a = -3, b = 5, and c = -2, find the value of a x b - b x c + a x c. [Integers]

- **(A)** 11
- **(B)** -1
- **(C)** -29
- **(D)** -19
- **(E)** 1
- **(F)** 21
- **(G)** -11

93. Acid plus a metal can give a salt and hydrogen gas. In an experiment, 5 of 8 test tubes show bubbles (a chemical change); the rest only get warm. What percent show a chemical change, and forming a new gas is which kind of change? [Acids, Bases & Salts]

- **(A)** 62.5%; no change
- **(B)** 50%; chemical
- **(C)** 62.5%; chemical
- **(D)** 60%; chemical
- **(E)** 62.5%; physical
- **(F)** 37.5%; chemical

94. A worker is paid Rs 300 per day plus Rs 20 for each item made beyond 10. Write the day's pay for making 10 + k items, and find the pay when k = 8. [Algebraic Expressions]

- **(A)** 300 + 20k; Rs 460
- **(B)** 300 + 20k; Rs 440
- **(C)** 300 + 20k; Rs 360
- **(D)** 300 + 20(10+k); Rs 660
- **(E)** 300 + 20k; Rs 560
- **(F)** 300k + 20; Rs 2420
- **(G)** 320 + 20k; Rs 480

95. An aeration tank's 'activated sludge' needs dissolved oxygen above 2 mg/L to keep microbes alive. A reading shows 1.4 mg/L. By how much must oxygen rise to reach a safe 3 mg/L, and low oxygen mainly harms what? [Wastewater Story]

- **(A)** 1.6 mg/L; the bar screens
- **(B)** 4.4 mg/L; the helpful microbes
- **(C)** 1.6 mg/L; the chlorine
- **(D)** 1.4 mg/L; the helpful microbes
- **(E)** 1.6 mg/L; the helpful microbes
- **(F)** 2 mg/L; the helpful microbes
- **(G)** 0.6 mg/L; the helpful microbes

96. Two divers, P and Q, are at -36 m and -52 m. P descends 20 m while Q rises 16 m. Who is deeper now, and by how many metres? [Integers]

- **(A)** P, by 4 m
- **(B)** Q, by 16 m
- **(C)** P, by 16 m
- **(D)** Q, by 4 m
- **(E)** P, by 20 m
- **(F)** Q, by 20 m
- **(G)** Both equal

97. An indicator chart scores acid = -1, neutral = 0, base = +1. A student tests vinegar, lime water, salt water, soap, and lemon juice, and adds up the scores. What is the total score? [Acids, Bases & Salts]

- **(A)** 2
- **(B)** 0
- **(C)** -2
- **(D)** -1
- **(E)** +1
- **(F)** +2
- **(G)** -3

98. Simplify 4x + 3y - 2x + 5 - y - 8, state how many terms remain, and give its value at x = 5, y = 4. [Algebraic Expressions]

- **(A)** 2x + 2y - 3; 3 terms; value 15
- **(B)** 2x + 2y - 3; 4 terms; value 15
- **(C)** 2x + 4y - 3; 3 terms; value 23
- **(D)** 2x + 2y + 13; 3 terms; value 31
- **(E)** 2x + 2y - 3; 3 terms; value 23
- **(F)** 2x + 2y - 3; 3 terms; value 11
- **(G)** 6x + 4y - 3; 3 terms; value 43

99. A town's plant recovers biogas that replaces 24 LPG cylinders' worth of fuel, each saving Rs 900. What total fuel cost is saved, and recovering biogas also reduces what? [Wastewater Story]

- **(A)** Rs 2,16,000; the waste sent to dumps
- **(B)** Rs 21,600; nothing useful
- **(C)** Rs 21,600; the number of people
- **(D)** Rs 21,600; the amount of clean water
- **(E)** Rs 21,600; the waste sent to dumps
- **(F)** Rs 18,000; the waste sent to dumps
- **(G)** Rs 9,000; the waste sent to dumps

100. Replace the star: (-9) x 6 + (star) = -40. What integer is the star? [Integers]

- **(A)** -94
- **(B)** -14
- **(C)** 4
- **(D)** 94
- **(E)** -4
- **(F)** 14